Module - 1

Intro to detection and incident response

to learn

1. Incident response lifecycle
2. Monitor and analyze network traffic
3. Process and Procedure
4. Interpret logs and alerts

The incident response lifecycle

incident handler's journal

incident response teams

Documentation, detection and managment tools

what is incident : An occurence that actually or imminently jeoperdize, without lawful authority, the confidentiality, Integrety or availability of information or information systems; or constitues a violation or immenent threat of violation of law security policies, security procedures or acceptable use policies.

Event :- An observable occurance on a network, system, or device

all security Incidents are events but not all event an incident

the 5 w of an incident

1. who triggerd the incident
2. what happened
3. when the Incident took place
4. where did it took place
5. why incident occured

Best way to do it is by using Incident handler's journel

-> A form of documentation used in Incident Response

Incident response teams

CSIRT (computer security Incident Response teams)

A specialised group of security professional that are trained in incident managment and response.

Roles in CSIRT

1. security analyst - Detection
2. Technical lead - leadership
3. Incident coordinator - Track and manage activites

Roles in Response

Teir 1 :- SOC Analyst

the first tier is composed of the least Experience SOC analyst also known as L1s

- Monitor, review, Prioritize alert based on criticality, severity
- Creating and closing alert using tickets
- Escalate ticket to L2, L3

Tier-2 SOC analyst

- More Experienced SOC
- Recieve escalated ticket from L1
- Configure and refine security tools
- Report to SOC lead

Tier 3 :- SOC lead

- Manage the operation of their team
- Explore methods of detection by advance detection Technique, such as malware and forensic analysis
- Reporting to SOC manager

Incident response Plans

- A document that outline the procedure to take in each step of incident Response

Elements of incident Plan

1. Incident response procedures
2. System Information
3. other Documents

Tools for Incident Response

Tools type

1. Detection & managment tools
2. Documentation tools
3. Investigative tools

Importance of documentation & types of Documentation

Documantation : Any form of recorded content that is used for a specific purpose

Types of Documents

1. Playbooks
2. Incident handlers journal
3. Policies
4. Plans
5. final report

Intrusion Detection

IDS is an application that monitor system and network activity and produce alerts on possible intrusion

IPS (Intrusion Prevention System) An application that monitor system activity for intrusion and take action to stop the activity

Popular IDS and IPS tools

- Snort
- Zeek
- Kismet
- Sagan
- Suricata

EDR Endpoint Detection an respon tools

1. Bitdefender end Point
2. Open EDR
3. FortiEDR

Alert and SIEM and SOAR Tools

Security Information and Event Managment (SIEM) Tool

∴ An Application that collects and analyzes log data to monitor critical activities in an Organisation

SIEM Process

Order ↓

1. Collection and aggregate data (logs)
2. Normalize data (Remove unwanted)
3. Analyze data

Security orchestration, automation and Response (SOAR)

∴ A collection of Application, tools, and workflow that uses automation to respond to security events.

Some of widley Known SIEM Tools

1. Alien Vault OSSIM
2. Chronical
3. Elastic
4. Exabeam
5. IBM Q Radar
6. Log Rythm
7. SplunR

Module - 2

Network monitoring and analysis

to learn

1. Network traffic flows
2. Packet Sniffers
3. Packet analysis

The importance of network traffic flows.

Network traffic : the amount of data that moves across a Network

Network data :- the data that's transmitted between devices on a Network

IOC (Indicators of compromise) :- observable evidence that suggest a potential security Incident.

1 Data Exfiltration :- Unauthorized transmission of data from a system

Maintaing Awarness of Network Monitoring

Know your Network.

Network communication Provide information about connection such as source and destination IP addresses, amount of data transferred, date and time. and more. This information can be valuable for security Professionals when developing baselines of Normal or Expected behaviour

Baseline is a refrence Point that's used for comparison.

Monitor your network.

Flow analysis

flow refers to the movement of network communication and includes information related to packets, Protocols and ports.

However, malicious actor can use protocols and ports that are not commonly associated to maintain communication between compromised system and themself
also called command and control (C2)

Packet Payload information

Temporal patterns

Defesive measures

1. Prevant attacker access
2. monitor network activity
3. Protect assets
4. Detect and stop the exfiltration

Capture and view network traffic.

More about Packet capture.

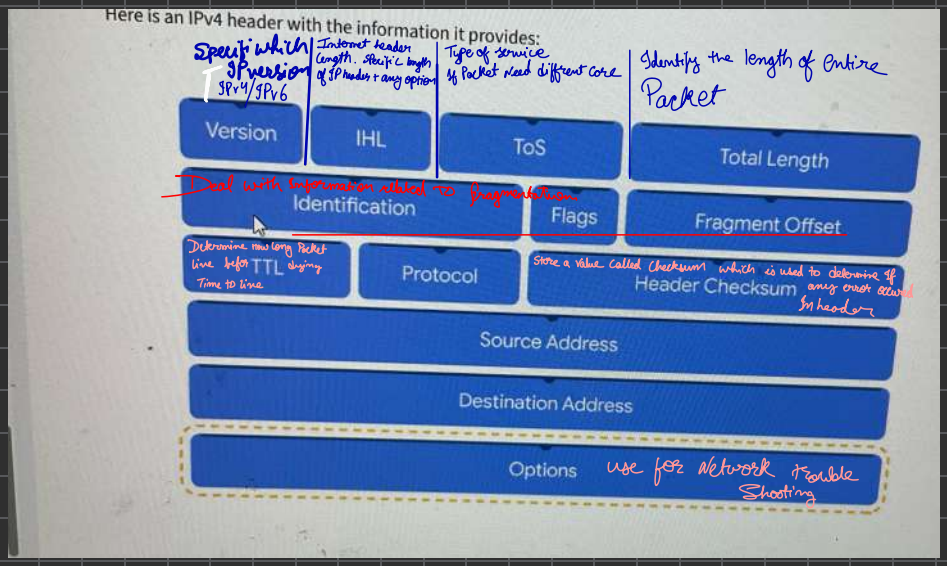
Data packet is basic unit information that travels from one device to another within network

Made of 3 components

Header Payload footer

Header :- Packets begin with the most essential components. the header

Packets can have several headers depending on the protocol used such as Ethernet header, an IP header a TCP header and more. Headers provide information that's used to route Packages. to destination



Payload :- component directly follow the header and contains the actual data being delivered

Footer :- located at the end of footer. the ethernet protocol uses footers to provide error-checking information to determine if data has been corrupted

Most protocol, such as the Internet Protocol (IP)

Network protocol analyzers

Network Protocol analyzer (Packet Sniffers) are tools designed to capture and analyze traffic within a network

Ex tcp dump, wireshark, TShark

How network protocol analyzers work

Network protocol analyzers use both software and hardware capabilities to capture network traffic and display it for security analysts to examine and analyze. Here's how:

1. First, packets must be collected from the network via the **Network Interface Card (NIC)**, which is hardware that connects computers to a network, like a router. NICs receive and transmit network traffic, but by default they only listen to network traffic that's addressed to them. To capture all network traffic that is sent over the network, a NIC must be switched to a mode that has access to all visible network data packets. In wireless interfaces this is often referred to as monitoring mode, and in other systems it may be called promiscuous mode. This mode enables the NIC to have access to all visible network data packets, but it won't help analysts access all packets across a network. A network protocol analyzer must be positioned in an appropriate network segment to access all traffic between different hosts.
2. The network protocol analyzer collects the network traffic in raw binary format. Binary format consists of 0s and 1s and is not as easy for humans to interpret. The network protocol analyzer takes the binary and converts it so that it's displayed in a human-readable format, so analysts can easily read and understand the information.

Note: Enabling promiscuous can expose your device to potential attacks because it allows sensitive information like passwords and other confidential data to be captured. It's important to use tools that operate in promiscuous mode responsibly and with caution.

Capturing packets

Packet sniffing is the practice of capturing and inspecting data packets across a network. A **packet capture (p-cap)** is a file containing data packets intercepted from an interface or network. Packet captures can be viewed and further analyzed using network protocol analyzers. For example, you can filter packet captures to only display information that's most relevant to your investigation, such as packets sent from a specific IP address.

Note: Using network protocol analyzers to intercept and examine private network communications without permission is considered illegal in many places.

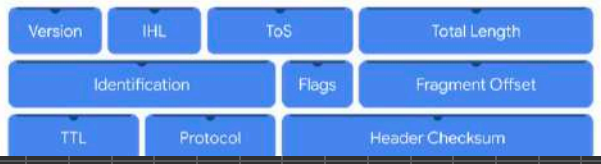
P-cap files can come in many formats depending on the packet capture library that's used. Each format has different uses and network tools may use or support specific packet capture file formats by default. You should be familiar with the following libraries and formats:

1. **Libpcap** is a packet capture library designed to be used by Unix-like systems, like Linux and MacOS®. Tools like tcpdump use Libpcap as the default packet capture file format.
2. **WinPcap** is an open-source packet capture library designed for devices running Windows operating systems. It's considered an older file format and isn't predominantly used.
3. **Npcap** is a library designed by the port scanning tool Nmap that is commonly used in Windows operating systems.
4. **PCAPng** is a modern file format that can simultaneously capture packets and store data. Its ability to do both explains the "ng," which stands for "next generation."

Pro tip: Analyzing your home network can be a good way to practice using these tools.

Deffrient Protocol contain diffrent headers

- **Internet Header Length (IHL)**: This field specifies the length of the IPv4 header including any Options.
- **Type of Service (ToS)**: This field provides information about packet priority for delivery.
- **Total Length**: This field specifies the total length of the entire IP packet including the head and the data.
- **Identification**: Packets that are too large to send are fragmented into smaller pieces. This field specifies a unique identifier for fragments of an original IP packet so that they can be reassembled once they reach their destination.
- **Flags**: This field provides information about packet fragmentation including whether the original packet has been fragmented and if there are more fragments in transit.
- **Fragment Offset**: This field is used to identify the correct sequence of fragments.
- **Time to Live (TTL)**: This field limits how long a packet can be circulated in a network, preventing packets from being forwarded by routers indefinitely.
- **Protocol**: This field specifies the protocol used for the data portion of the packet.
- **Header Checksum**: This field specifies a checksum value which is used for error-checking the header.
- **Source Address**: This field specifies the source address of the sender.
- **Destination Address**: This field specifies the destination address of the receiver.
- **Options**: This field is optional and can be used to apply security options to a packet.



IPv6

IPv6 adoption has been increasing because of its large address space. There are eight fields in the header:

- **Version**: This field indicates the IP version. For an IPv6 header, IPv6 is used.
- **Traffic Class**: This field is similar to the IPv4 Type of Service field. The Traffic Class field provides information about the packet's priority or class to help with packet delivery.
- **Flow Label**: This field identifies the packets of a flow. A flow is the sequence of packets sent from a specific source.
- **Payload Length**: This field specifies the length of the data portion of the packet.
- **Next Header**: This field indicates the type of header that follows the IPv6 header such as TCP.
- **Hop Limit**: This field is similar to the IPv4 Time to Live field. The Hop Limit limits how long a packet can travel in a network before being discarded.
- **Source Address**: This field specifies the source address of the sender.
- **Destination Address**: This field specifies the destination address of the receiver.

Wireshark

Wireshark is an open-source network protocol analyzer. It uses a graphical user interface (GUI), which makes it easier to visualize network communications for packet analysis purposes. Wireshark has many features to explore that are beyond the scope of this course. You'll focus on how to use basic filtering to isolate network packets so that you can find what you need.

Display filters

Wireshark's display filters let you apply filters to packet capture files. This is helpful when you are inspecting packet captures with large volumes of information. Display filters will help you find specific information that's most relevant to your investigation. You can filter packets based on information such as protocols, IP addresses, ports, and virtually any other property found in a packet. Here, you'll focus on display filtering syntax and filtering for protocols, IP addresses, and ports.

Comparison operators

You can use different comparison operators to locate specific header fields and values. Comparison operators can be expressed using either abbreviations or symbols. For example, this filter using the `==` equal symbol in this filter `ip.src == 8.8.8.8` is identical to using the `eq` abbreviation in this filter `ip.src eq 8.8.8.8`.

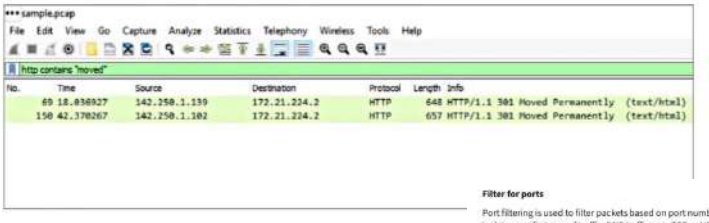
This table summarizes the different types of comparison operators you can use for display filtering.

Operator type	Symbol	Abbreviation
Equal	`==`	eq
Not equal	`!=`	ne
Greater than	`>`	gt
Less than	`<`	lt
Greater than or equal to	`>=`	ge
Less than or equal to	`<=`	le

Pro tip: You can combine comparison operators with Boolean logical operators like `and` and `or` to create complex display filters. Parentheses can also be used to group expressions and to prioritize search terms.

Contains operator

The `contains` operator is used to filter packets that contain an exact match of a string of text. Here is an example of a filter that displays all HTTP streams that match the keyword `"moved"`.

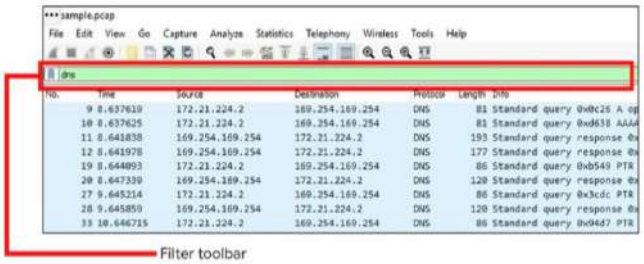


Matches operator

The `matches` operator is used to filter packets based on the regular expression (regex) that's specified. Regular expression is a sequence of characters that forms a pattern. You'll explore more about regular expressions later in this program.

Filter toolbar

You can apply filters to a packet capture using Wireshark's filter toolbar. In this example, `dns` is the applied filter, which means Wireshark will only display packets containing the DNS protocol.



Pro tip: Wireshark uses different colors to represent protocols. You can customize colors and create your own filters.

Filter for protocols

Protocol filtering is one of the simplest ways you can use display filters. You can simply enter the name of the protocol to filter. For example, to filter for DNS packets simply type `dns` in the filter toolbar. Here is a list of some protocols you can filter for:

- dns
- http
- ftp
- ssh
- arp
- telnet
- icmp

Filter for an IP address

You can use display filters to locate packets with a specific IP address.

For example, if you would like to filter packets that contain a specific IP address use `ip.addr`, followed by a space, the equal `==` comparison operator, and the IP address. Here is an example of a display filter that filters for the IP address `172.21.224.2`:

```
ip.addr == 172.21.224.2
```

To filter for packets originating from a specific source IP address, you can use the `ip.src` filter. Here is an example that looks for the `10.10.10.10` source IP address:

```
ip.src == 10.10.10.10
```

To filter for packets delivered to a specific destination IP address, you can use the `ip.dst` filter. Here is an example that searches for the `4.4.4.4` destination IP address:

```
ip.dst == 4.4.4.4
```

Filter for a MAC address

You can also filter packets according to the **Media Access Control (MAC) address**. As a refresher, a MAC address is a unique alphanumeric identifier that is assigned to each physical device on a network.

Here's an example:

```
eth.addr == 00:70:f4:23:18:c4
```

Filter for ports

Port filtering is used to filter packets based on port numbers. This is helpful when you want to isolate specific types of traffic. DNS traffic uses TCP or UDP port 53 so this will list traffic related to DNS queries and responses only.

For example, if you would like to filter for a UDP port:

```
udp.port == 53
```

Likewise, you can filter for TCP ports as well:

```
tcp.port == 25
```

Follow streams

Wireshark provides a feature that lets you filter for packets specific to a protocol and view streams. A stream or conversation is the exchange of data between devices using a protocol. Wireshark reassembles the data that was transferred in the stream in a way that's simple to read.

Following a protocol stream is useful when trying to understand the details of a conversation. For example, you can examine the details of an HTTP conversation to view the content of the exchanged request and response messages.